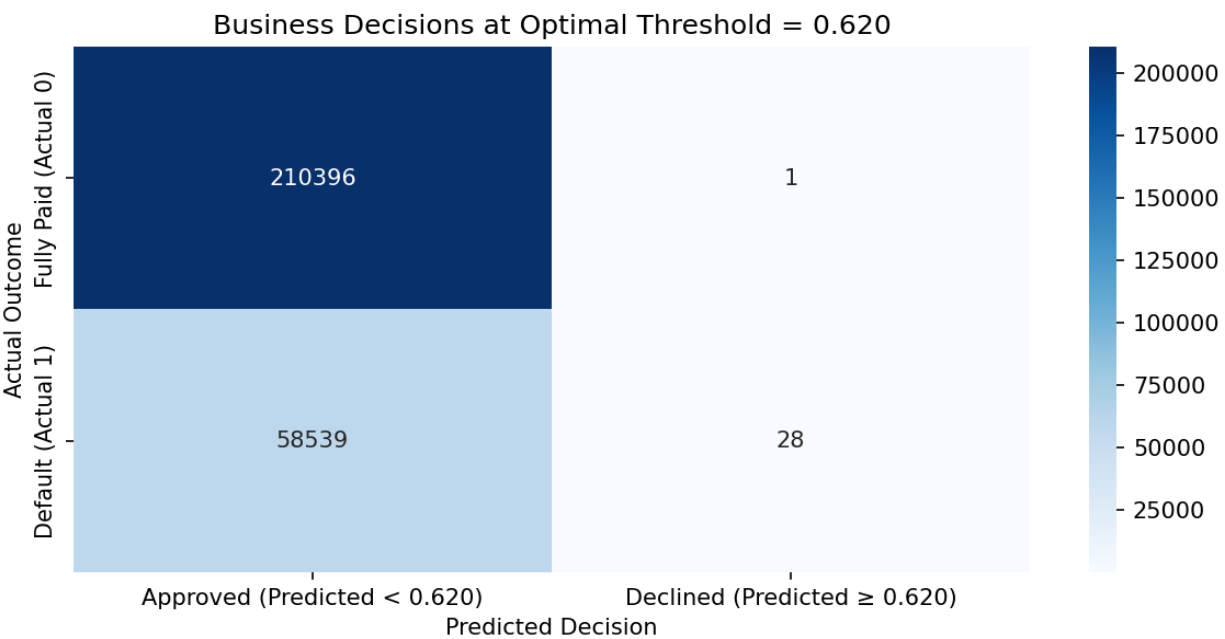


Profit-Driven Credit Approval Thresholding

From Predictions to Portfolio Outcomes

Optimal Threshold: 0.620 | Generated \$123,795,944 Net Cash Flow

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Data Science Project
June 19, 2026



Total Loans: 268,964
Approved: 268,935
Declined: 29

Figure 1: Approval decisions at the optimal threshold (0.620).

Methodology & Validation

The Pipeline

Using 1.34M LendingClub loans (2007–2018), I built a temporal validation pipeline:

- **Data:** Filtered resolved loans; binary default target.
- **Profit Metric:** $\text{Net Cash Flow} = \text{total_received} - \text{funded_amount}$.
- **Split:** 80/20 temporal split by issue date (Train: 2007–2015, Test: 2016–2018).
- **Engineering:** 4 feature plans with `SafeFeatureEngineer` to prevent leakage.
- **Optimization:** Swept 90 thresholds (0.05–0.95) to maximize profit.

The AUC Paradox

In this experiment, the model with higher AUC (Random Forest, 0.7001) did not produce higher profit than the lower-AUC model (Logistic Regression, 0.6671).

Why? The Logistic Regression produced a sharper profit peak at the optimal threshold (0.620). The Random Forest's probability estimates were less precise at the decision boundary, resulting in a flatter profit curve.

Observation: Calibration and threshold behavior can be as important as raw discriminative power when the objective is profit.

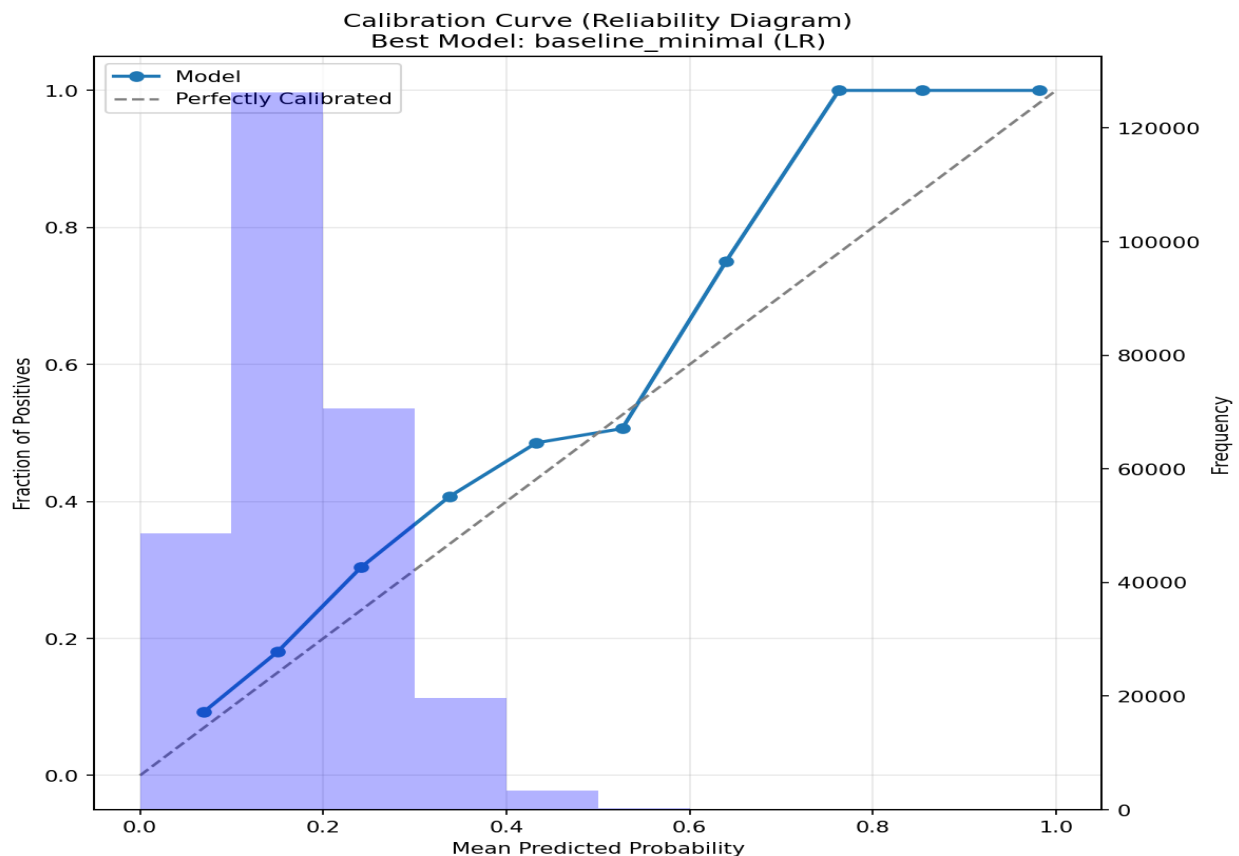


Figure 2: Calibration curve. The model's probabilities align with observed default rates.

Results: Model Performance & Baselines

Model Comparison

Plan	Model	AUC	Threshold	Profit
Baseline Minimal	LR	0.6671	0.620	\$123,795,944
Baseline Minimal	RF	0.7001	0.920	\$123,790,408
Domain Enhanced	LR	0.6524	0.890	\$123,790,408
Domain Enhanced	RF	0.6997	0.870	\$123,790,408
Interaction Heavy	LR	0.6805	0.940	\$123,679,150
Interaction Heavy	RF	0.7002	0.910	\$123,790,408
ML Informed	LR	0.6794	0.930	\$123,677,627
ML Informed	RF	0.6999	0.790	\$123,790,408

Baseline Validation

Strategy	Net Cash Flow	Outcome
Approve-All	\$-331,903,078	Loss
Optimal Threshold (0.620)	\$-331,622,596	Loss Reduced by \$280,481

Note: The test period (2016–2018) had negative returns. The threshold reduced losses by \$280k.

Limitations & Validation

Data Integrity: Leakage Detection

Initial models achieved **AUC 0.9999**, indicating target leakage. I identified 25 payment/collection columns and removed them using a pattern-based function. After removal, AUC dropped to **0.6671**, confirming realistic model performance.

Limitations

This is a retrospective simulation using historical data. The following factors were not modeled:

- Operational costs (underwriting, collections, capital costs).
- Dynamic economic shifts (the optimal threshold is conditional on historical distributions).
- Live deployment constraints (simulation vs. real-world execution).

Potential Extensions

- **Dynamic Thresholding:** Adjust the cut-off based on macro-economic indicators.
- **Profit-Weighted Training:** Directly optimize for financial outcomes during training.
- **Multi-Period Optimization:** Model portfolio dynamics over time.

Full technical code and 15-page academic report available on [GitHub](#).